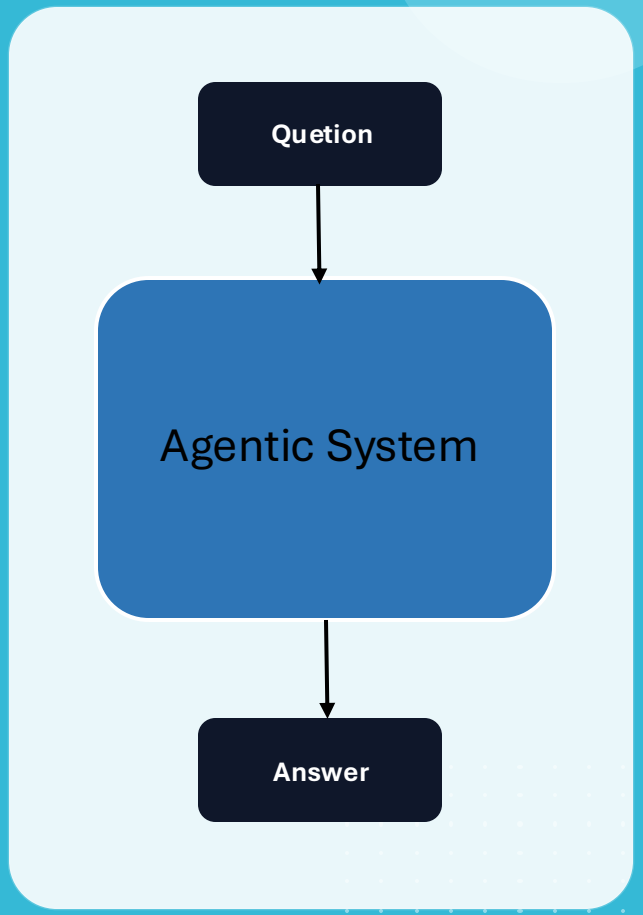
VideoSearcher

Vision-Language Interaction

Extending an agentic video search system with visual grounding and cross-modal reasoning



Cloud Computing: From Enterprise Application to AI@Edge - Class Project

Garzoni - Guarnizo - Scappatura

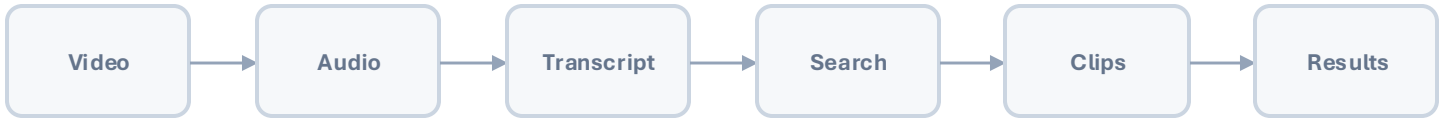
BASELINE

Starting point: the inherited system

What was already there

- Agentic orchestration
- Audio extraction
- Whisper transcription
- Transcript search
- Dockerized MCP tools

Inherited audio/text pipeline



Gap we addressed

The system could search what was said, but could not reason about what was shown.

PROJECT FOCUS

Problem and objectives

Upgrade VideoSearcher from transcript search to video understanding.

01

Add visual grounding

Answer questions about objects, scenes and frame content.

02

Connect modalities

Use spoken phrases to locate moments and inspect their frames.

03

Keep it modular

Add vision without breaking the inherited MCP architecture.

04

Make it testable

Evaluate multi-step behavior across prompt specificity levels.

APPENDIX

What we added

Extension of the system from all points of view!

Orchestrator Models

8 (+6)

Tools

12 (+6)

Evaluation metrics

14 (+11)

VQA

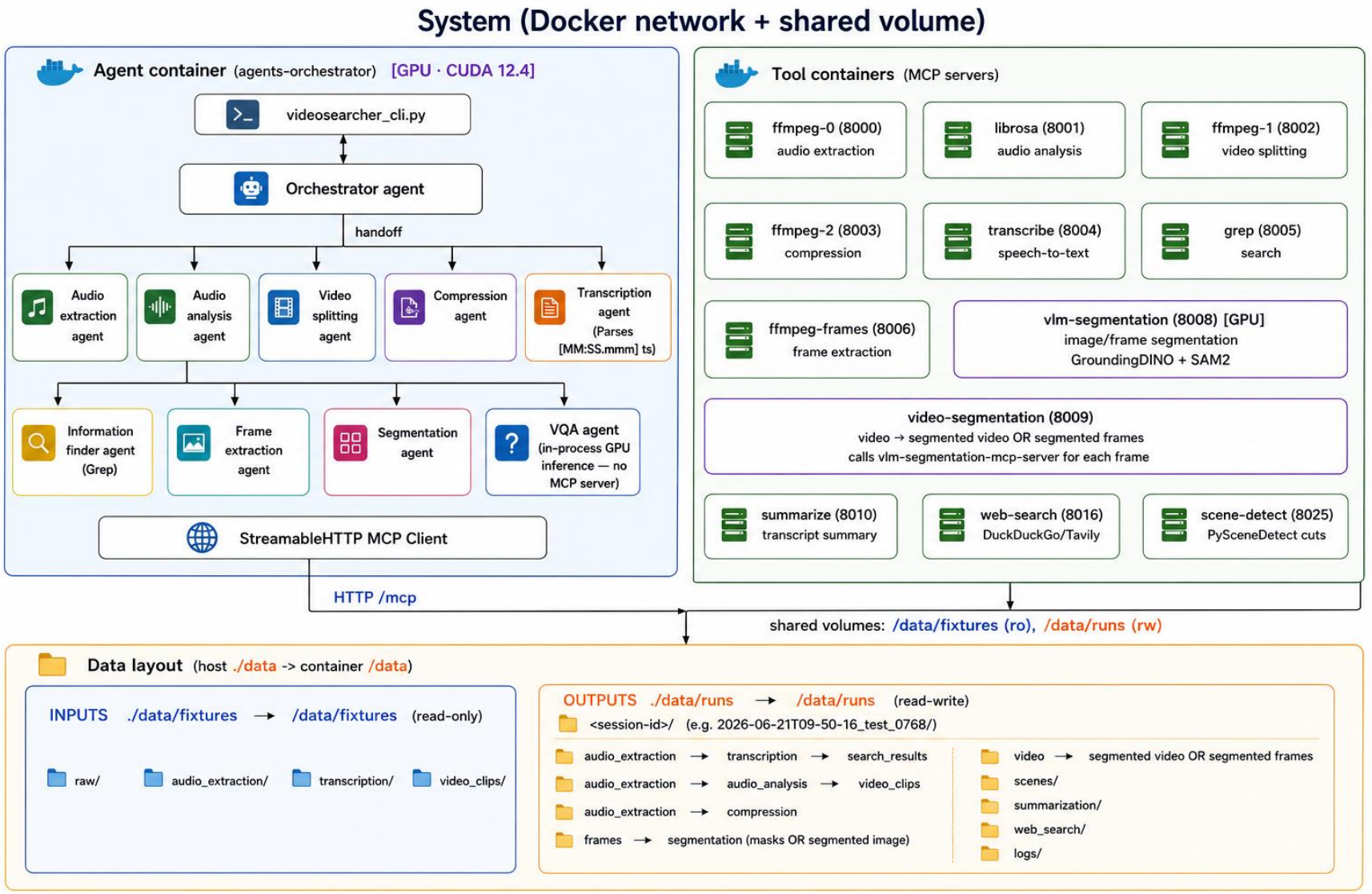
Segmentation

Scenarios

22 (+13)

BASELINE

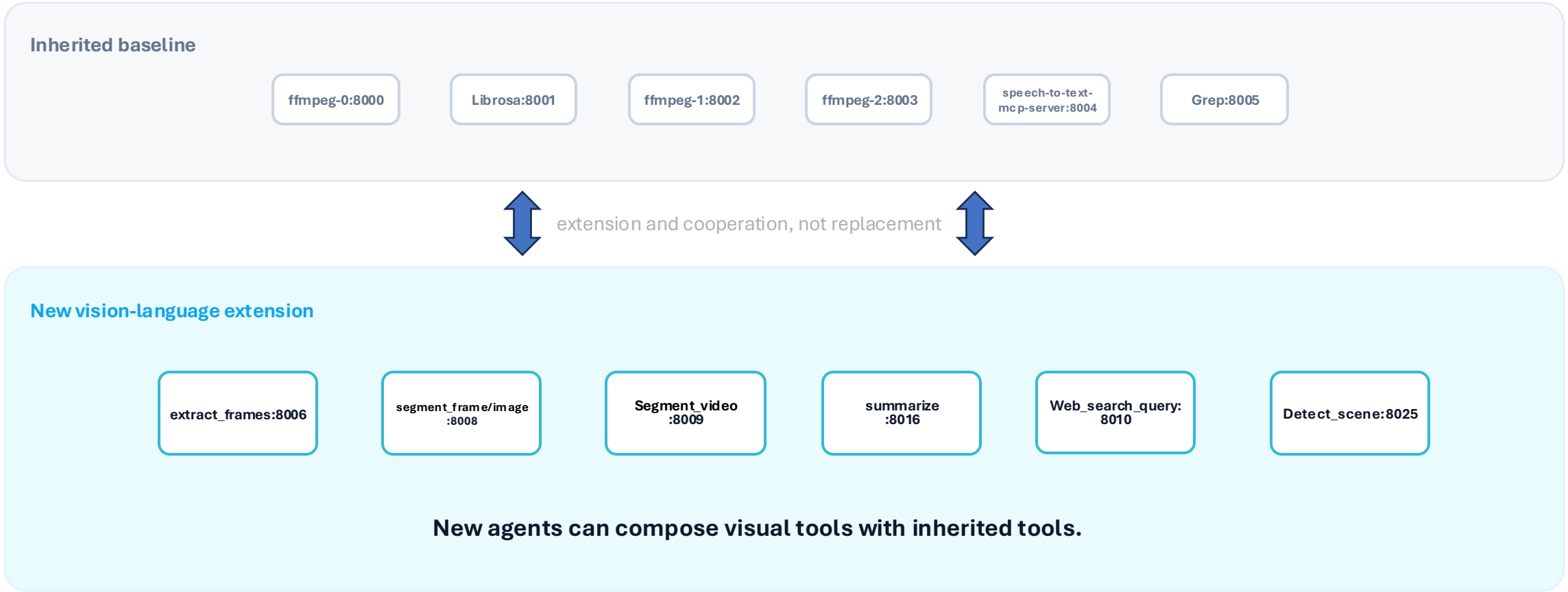
Complete System Architecture



SYSTEM DESIGN

New tools and application context layer extension

audio-and-text pipeline to visual capabilities



New tools added and application context extensions

VQA DEEP DIVE

GPU · IN-PROCESS

VQA engine: in-process visual reasoning

Phi-4 Vision turns extracted frames into natural-language evidence without a separate MCP server.



1. Input contract

- frame_path + natural-language question
- accepts vague prompts such as “what is visible?”
- uses the actual extracted frame, not the transcript alone

2. Runtime behavior

- opens the image and converts it to RGB
- lazy-loads Phi-4 Vision on first use
- builds an <image> + question prompt
- reuses the GPU-resident model afterwards

3. System role

- answers direct image/frame questions
- completes speech → timestamp → frame chains
- returns answer + model metadata
- keeps heavy visual reasoning in the core container

Runtime output

{ success, frame_path, question, answer, model }

Why this design matters

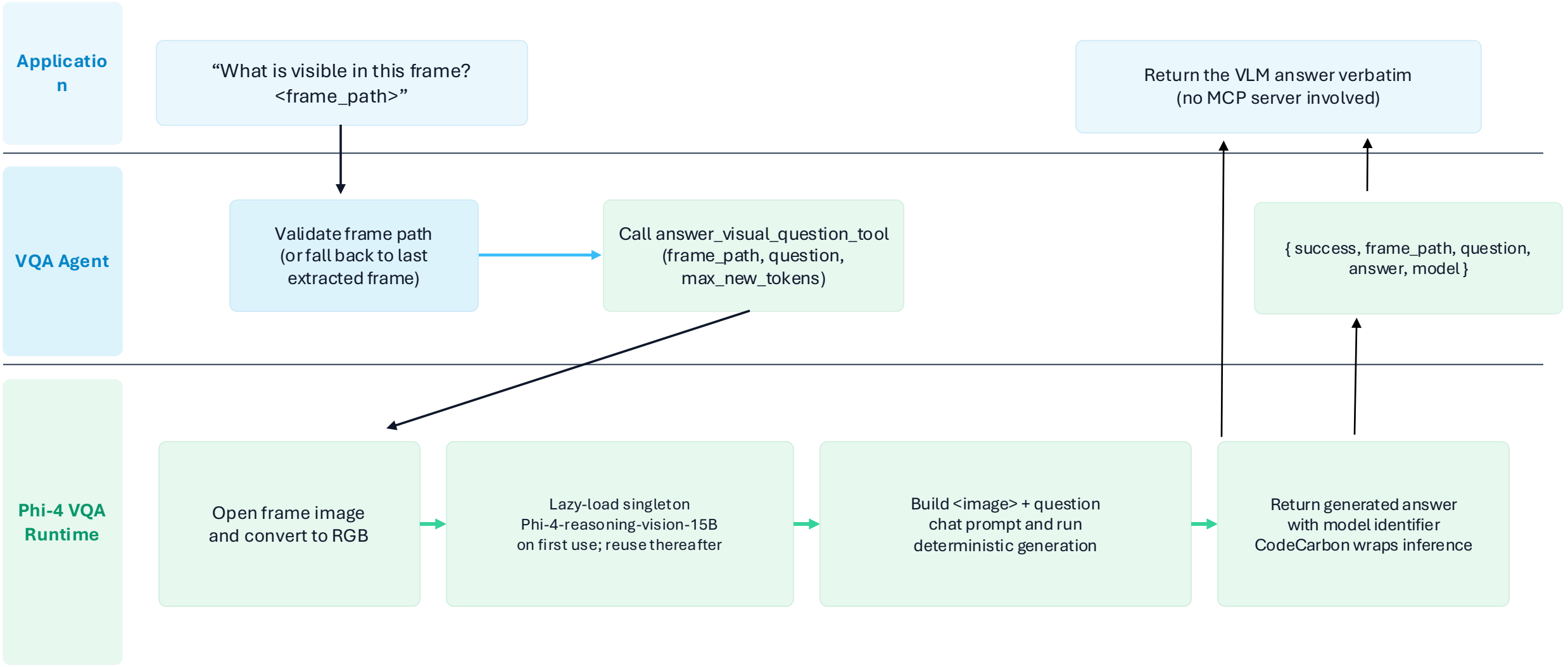
Gemma plans and routes; Phi-4 Vision interprets pixels. VQA is exposed as a handoff, but the model inference is not an MCP microservice.

Key takeaway: VQA is the visual reasoning endpoint of the pipeline — it turns an extracted frame or an image into an open-ended answer in image context.

VQA - Verified In-Process Flow

IN-PROCESS

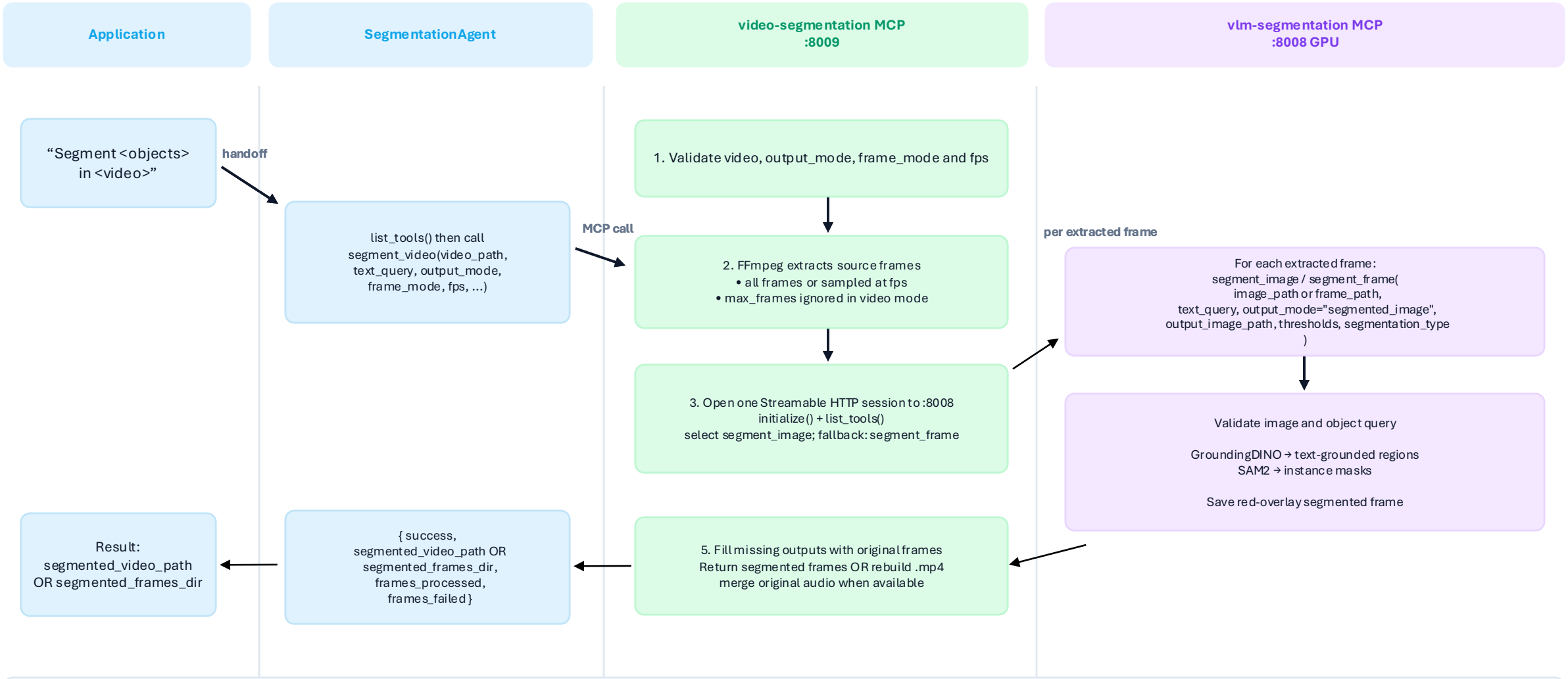
VqaAgent runs inside agents-orchestrator | Function tool: answer_visual_question_tool(frame_path, question)



video-segmentation-mcp-server

Port: 8009

SegmentationAgent → video-segmentation-mcp-server :8009 → vlm-segmentation-mcp-server :8008 GPU

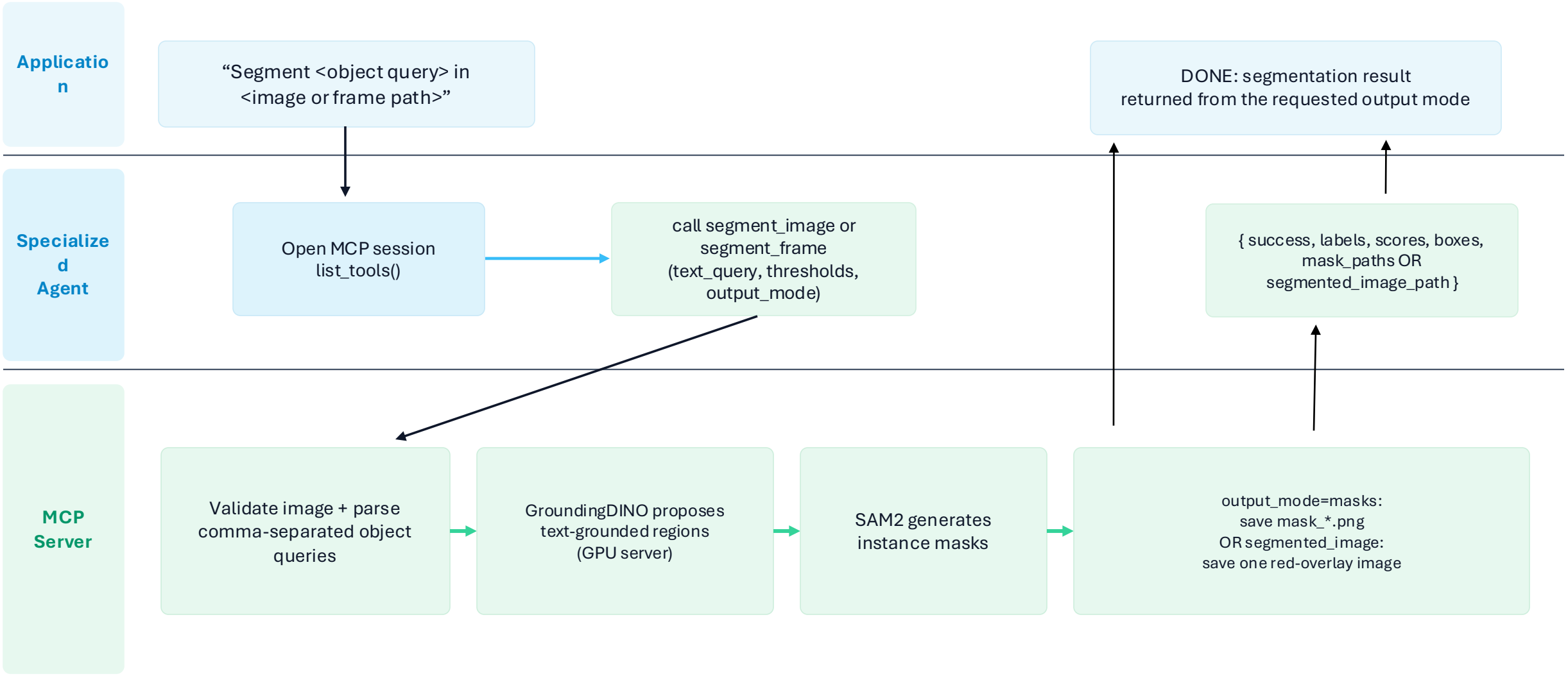


Architectural choice: :8009 is a lightweight video orchestration server. It delegates every extracted frame to the single GPU segmentation service (:8008), avoiding duplicate model loading.

vlm-segmentation-mcp-server

Port :8008

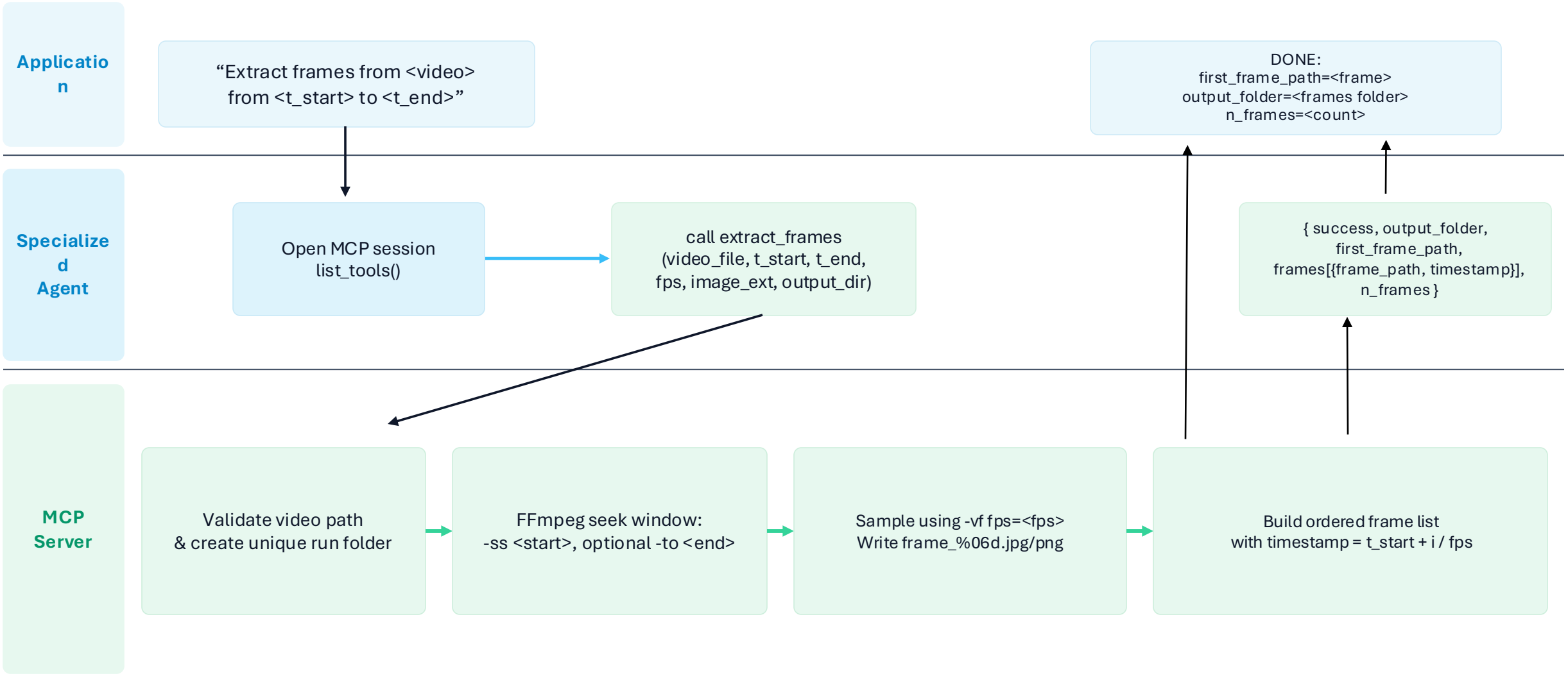
SegmentationAgent -> vlm-segmentation-mcp-server :8008 | Handlers: segment_image / segment_frame



ffmpeg-frames-mcp-server

Port: 8006

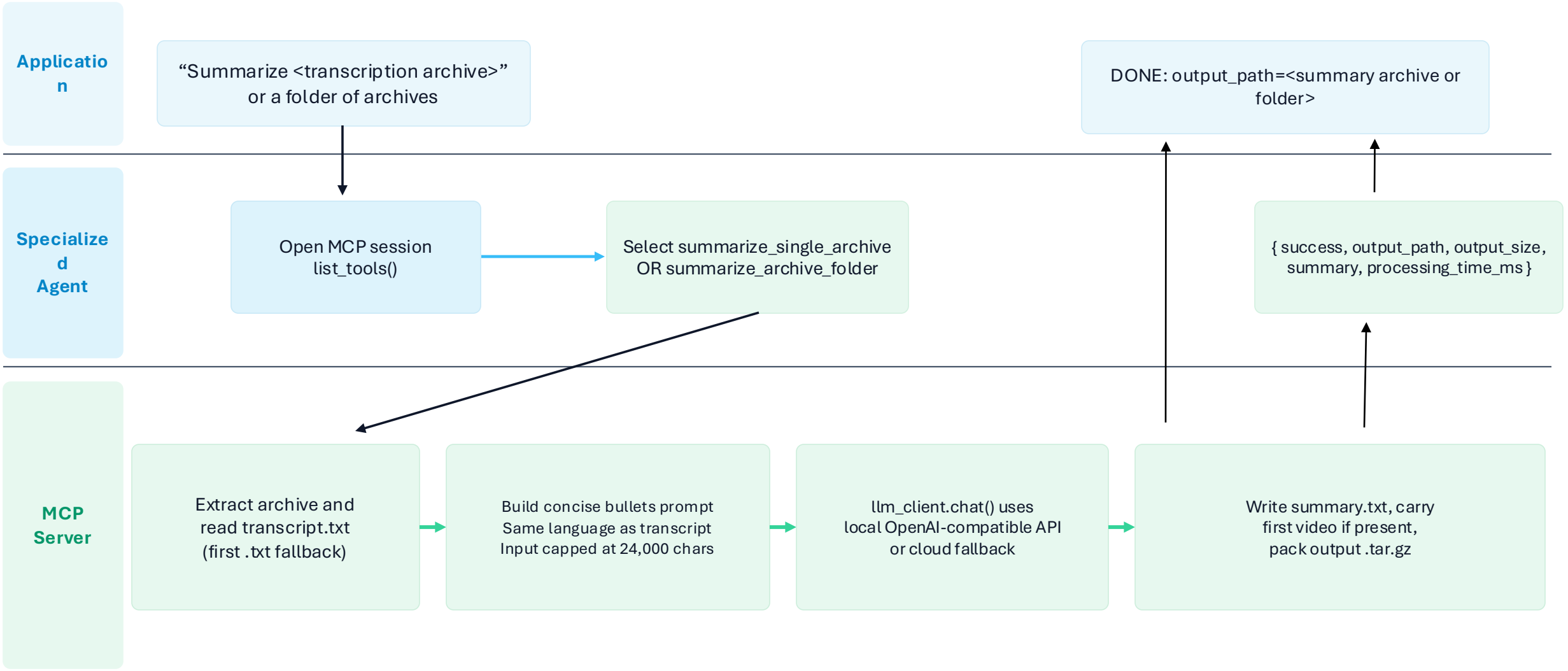
FrameExtractionAgent -> ffmpeg-frames-mcp-server :8006 | Handler: extract_frames(video_file, t_start, t_end, fps)



summarize-mcp-server

Port: 8016

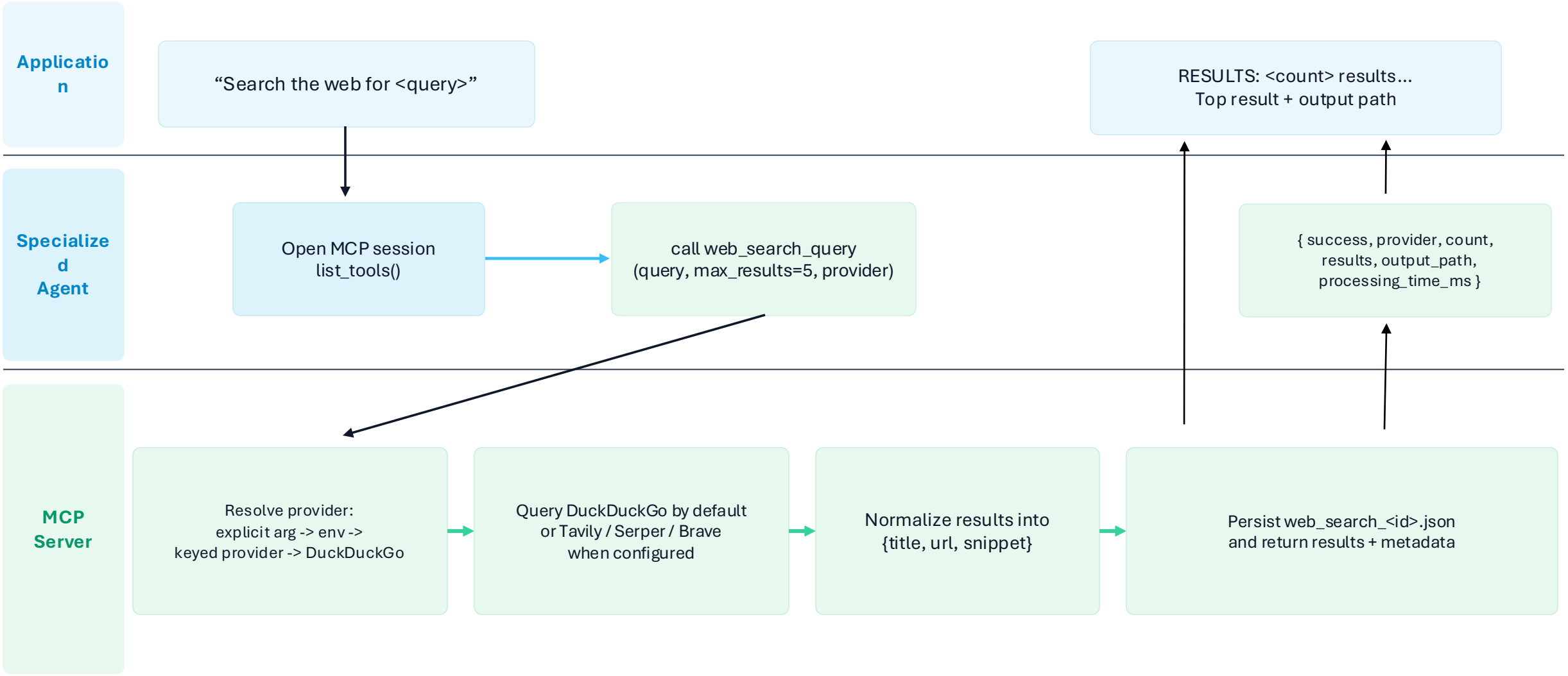
SummarizationAgent -> summarize-mcp-server :8016 | Handlers: summarize_single_archive / summarize_archive_folder



web-search-mcp-server

Port: 8010

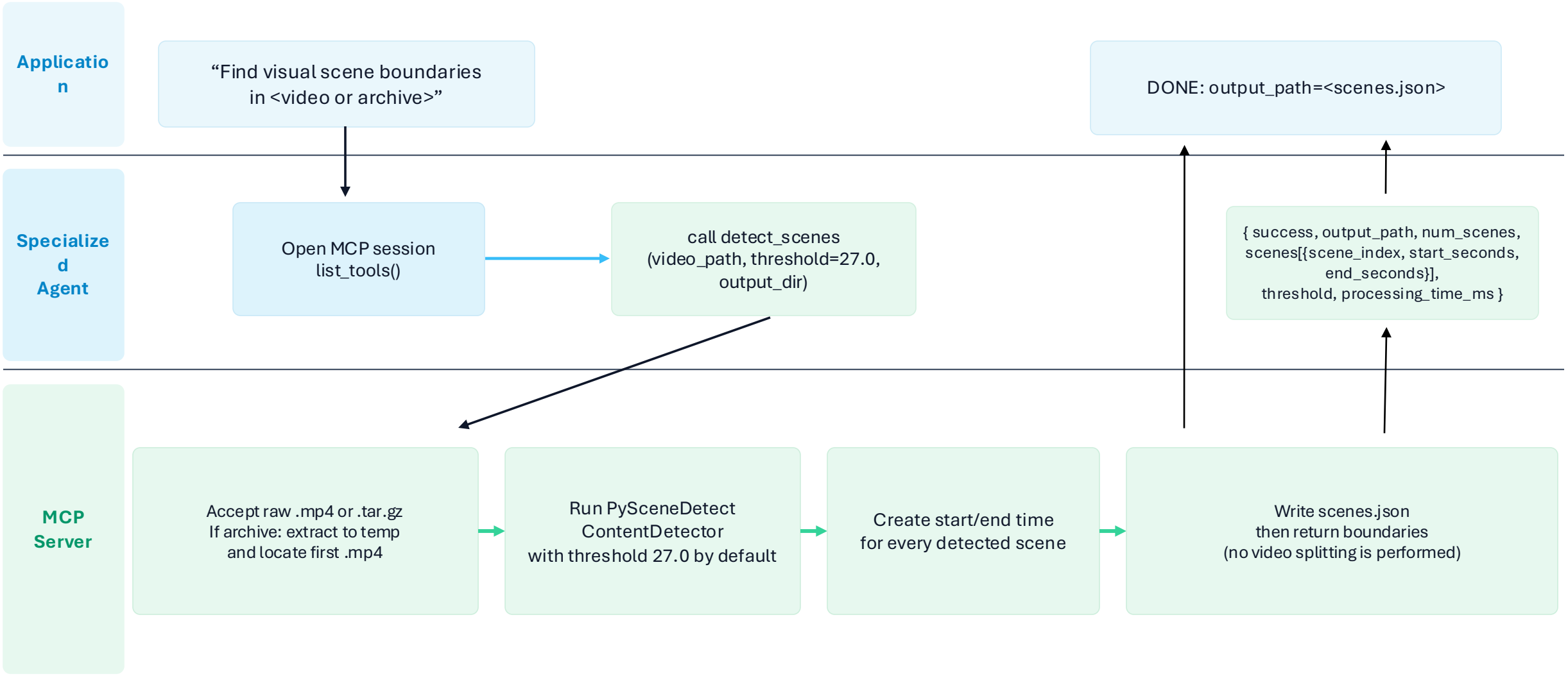
WebSearchAgent -> web-search-mcp-server :8010 | Handler: web_search_query(query, max_results, provider)



scene-detect-mcp-server

Port:8025

SceneDetectAgent -> scene-detect-mcp-server :8025 | Handler: detect_scenes(video_path, threshold)



VISION PIPELINE EXAMPLE

Visual pipeline: text-guided segmentation

Open-vocabulary object grounding: the user names what he want to segment
The system finds and masks/segment it



Output example: segmented image with overlay

1 **Frame / image**
Input path from video or user upload

2 **Text query**
person, car, lion, object...

3 **GroundingDINO**
Open-vocabulary box detection

4 **SAM2**
Pixel-level mask generation

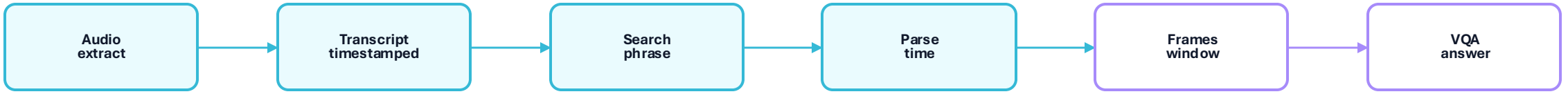
Grounded SAM2 style approach

Two output modes

masks for internal chains · segmented_image / video for final user-facing artifacts

Cross-modal reasoning: speech → timestamp → frame → answer

The system can locate a spoken moment and inspect the matching visual evidence.



Prompt example

“What is visible when the speaker mentions the F-35?”



VQA output

Answer based on the extracted frame, not only in transcript text.

This turns timestamped transcription into a bridge between language and pixels.

APPENDIX

Test scenario catalogue and metrics

Documented test set: 22 scenarios x 3 prompt levels = 66 core .

SC-01 - SC-09	Audio / text	9
SC-28 .. SC-37	Vision	10
SC-40 - SC-42 - SC-47	Extension	3

Prompt levels	vague · clear · algorithmic
Main checks	handoff sequence · result
Vision check	keywords · min length · output path

APPENDIX

Audio / text

Scenario	Difficulty	Scenario Name	Prompt Level	Input Prompt
SC-01	1 (Simple)	Single Tool: Audio Extraction	Vague	Extract the sound from this video. File: /data/fixtures/raw/sample_video.mp4
SC-01	1 (Simple)	Single Tool: Audio Extraction	Clear	Use Audio Extraction Agent to extract audio.wav from the video file at /data/fixtures/raw/sample_video.mp4.
SC-01	1 (Simple)	Single Tool: Audio Extraction	Algorithmic	Run audio_extraction_handoff. Input: /data/fixtures/raw/sample_video.mp4.
SC-02	2 (Simple)	Single Tool: Transcription	Vague	Turn this recording into text. File: /data/fixtures/audio_extraction/sample_archive.tar.gz
SC-02	2 (Simple)	Single Tool: Transcription	Clear	Run the transcription agent on the archive at /data/fixtures/audio_extraction/sample_archive.tar.gz.
SC-02	2 (Simple)	Single Tool: Transcription	Algorithmic	Run the transcription agent with the provided input archive: /data/fixtures/audio_extraction/sample_archive.tar.gz.
SC-03	3 (Simple)	Single Tool: Grep Search	Vague	Does this file mention 'elimination'? File: /data/fixtures/transcription/sample_transcript.tar.gz
SC-03	3 (Simple)	Single Tool: Grep Search	Clear	Search the transcript.txt in this archive for the word 'elimination'. Archive: /data/fixtures/transcription/sample_transcript.tar.gz
SC-03	3 (Simple)	Single Tool: Grep Search	Algorithmic	Call information finder agent with (search_pattern='elimination', archive_path=/data/fixtures/transcription/sample_transcript.tar.gz).
SC-04	4 (Moderate)	Early Pipeline (Initial Processing)	Vague	Find the sentence breaks in this video. File: /data/fixtures/raw/sample_video.mp4
SC-04	4 (Moderate)	Early Pipeline (Initial Processing)	Clear	Extract the audio and then find timestamps from /data/fixtures/raw/sample_video.mp4.
SC-04	4 (Moderate)	Early Pipeline (Initial Processing)	Algorithmic	1. Audio Extraction Agent on /data/fixtures/raw/sample_video.mp4 -> 2. Audio Analysis Agent. Output timestamps.txt in archive.
SC-05	5 (Moderate)	Late Pipeline (Text Analysis)	Vague	Find the word 'last week' in these video clips. Folder: /data/fixtures/video_clips/sample_clips
SC-05	5 (Moderate)	Late Pipeline (Text Analysis)	Clear	Compress the clips in /data/fixtures/video_clips/sample_clips, transcribe them, and search for 'last week'.
SC-05	5 (Moderate)	Late Pipeline (Text Analysis)	Algorithmic	Use following pipeline: Compression -> Transcription -> Information Finder with (query='last week') for folder /data/fixtures/video_clips/sample_clips.

APPENDIX

Vision

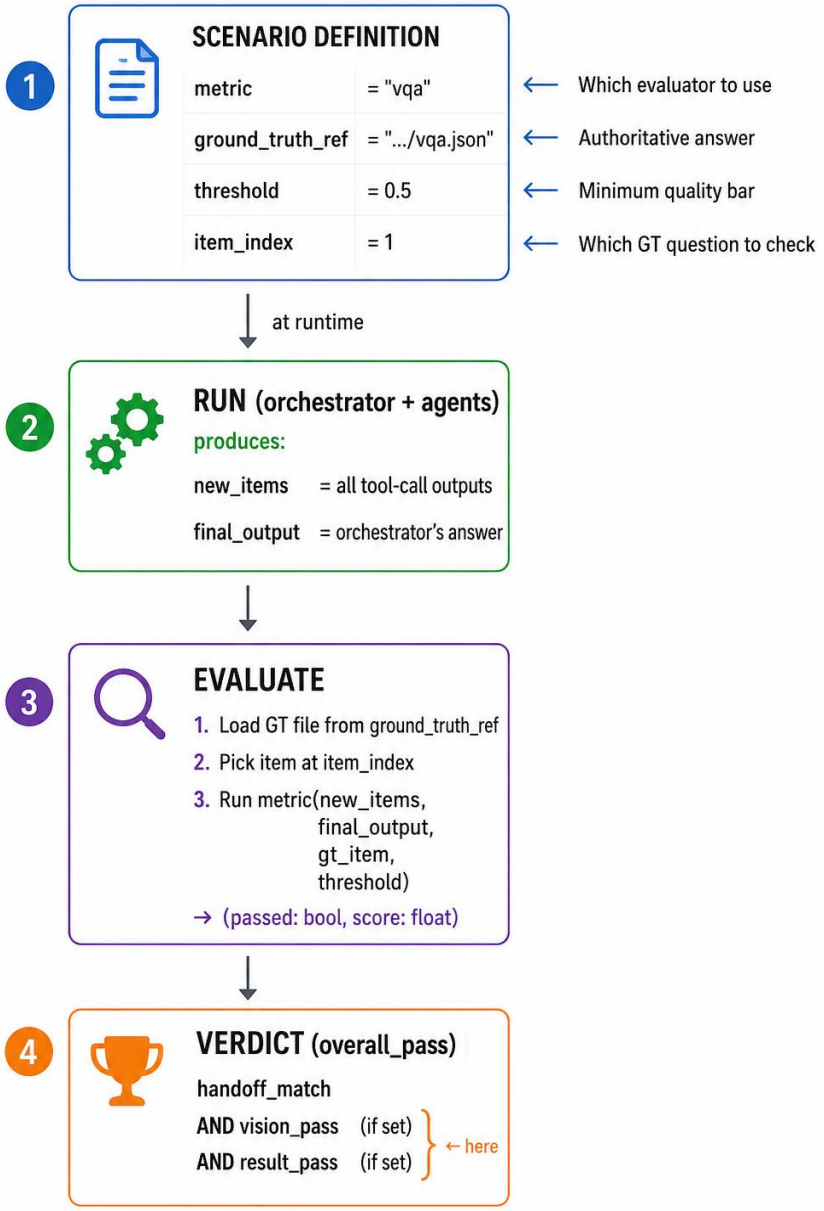
Scenario	Difficulty	Scenario Name	Prompt Level	Input Prompt
SC-28	2 (Simple)	Single Agent: Frame Extraction	Vague	Pull some images from this video at /data/fixtures/raw/lionandgorilla.mp4.
SC-28	2 (Simple)	Single Agent: Frame Extraction	Clear	Use the frame extraction agent to extract frames from the video at /data/fixtures/raw/lionandgorilla.mp4 between 0 and 5 seconds at 1 fps.
SC-28	2 (Simple)	Single Agent: Frame Extraction	Algorithmic	extract_frames_handoff. video_path=/data/fixtures/raw/lionandgorilla.mp4. t_start=0. t_end=5. fps=1.
SC-29	2 (Simple)	Single Agent: VQA on Image	Vague	What is shown in this picture at /data/fixtures/raw/lionandgorilla_frame0.jpg?
SC-29	2 (Simple)	Single Agent: VQA on Image	Clear	Use the VQA agent to answer: what animals appear in the image at /data/fixtures/raw/lionandgorilla_frame0.jpg?
SC-29	2 (Simple)	Single Agent: VQA on Image	Algorithmic	vqa_handoff. frame_path=/data/fixtures/raw/lionandgorilla_frame0.jpg. question='What animals are visible in this image?'
SC-30	3 (Simple)	2-Step Pipeline: Frame Extraction → VQA	Vague	Look at a frame from /data/fixtures/raw/lionandgorilla.mp4 and tell me what you see.
SC-30	3 (Simple)	2-Step Pipeline: Frame Extraction → VQA	Clear	Extract a frame from the video at /data/fixtures/raw/lionandgorilla.mp4 (around 2 seconds) and then use VQA to describe what animals are visible.
SC-30	3 (Simple)	2-Step Pipeline: Frame Extraction → VQA	Algorithmic	extract_frames_handoff(video_path=/data/fixtures/raw/lionandgorilla.mp4, t_start=1, t_end=3, fps=1) → vqa_handoff(frame_path=<extracted_frame>, question='What animals are in this image?')
SC-31	3 (Simple)	Single Agent: Image Segmentation	Vague	Segment the objects in /data/fixtures/raw/lionandgorilla_frame0.jpg and show me where they are.
SC-31	3 (Simple)	Single Agent: Image Segmentation	Clear	Use the segmentation agent to detect the animals in the image at /data/fixtures/raw/lionandgorilla_frame0.jpg. Return the detected bounding boxes and labels.
SC-31	3 (Simple)	Single Agent: Image Segmentation	Algorithmic	segmentation_handoff. image_path=/data/fixtures/raw/lionandgorilla_frame0.jpg. text_query='animal'. output_mode='masks'.

APPENDIX

Extension

Scenario	Difficulty	Scenario Name	Prompt Level	Input Prompt
SC-40	3 (Simple)	Single Tool: Summarize	Vague	Give me a short summary of this. File: /data/fixtures/transcription/sample_timestamped_transcript.tar.gz
SC-40	3 (Simple)	Single Tool: Summarize	Clear	Summarize the transcript in the archive at /data/fixtures/transcription/sample_timestamped_transcript.tar.gz.
SC-40	3 (Simple)	Single Tool: Summarize	Algorithmic	Run summarize_handoff with the input archive: /data/fixtures/transcription/sample_timestamped_transcript.tar.gz.
SC-42	2 (Simple)	Single Tool: Web Search	Vague	Look up the specifications of the F-35 online.
SC-42	2 (Simple)	Single Tool: Web Search	Clear	Use the web search agent to search the web for 'F-35 specifications'.
SC-42	2 (Simple)	Single Tool: Web Search	Algorithmic	web_search_handoff. query='F-35 specifications'. max_results=5.
SC-47	3 (Simple)	Single Tool: Scene Detection	Vague	Split this video by scene, not just by silence. File: /data/fixtures/raw/f-35.mp4
SC-47	3 (Simple)	Single Tool: Scene Detection	Clear	Detect the scene boundaries in the video /data/fixtures/raw/f-35.mp4.
SC-47	3 (Simple)	Single Tool: Scene Detection	Algorithmic	scene_detect_handoff. video_path=/data/fixtures/raw/f-35.mp4. threshold=27.0.

Check Result



APPENDIX

What answer each Metric?

Metric	Plain question	Pass condition
WER	How many words in the transcript are wrong?	≤ 30% error rate
grep	Did the system find all keyword matches?	Found 100% of expected occurrences
VQA	Did the answer mention the right facts about the image?	≥ 50% of required facts present
IoU	Do the detected boxes overlap well with the correct boxes?	Overlap ≥ 30% per object
summary	Did the summary cover the key points?	≥ 60% of key points mentioned
scene	Were scene cuts found at the right times?	≥ 70% of cuts within ±1.5 s
audio	Is the speech segment count close to the reference?	Count differs by ≤ 3

APPENDIX

Robustness by prompt level

Tests verify orchestration behavior across prompt specificity levels.

	Vague	Clear	Algorithmic
Audio / text	harder	checked	checked
Vision	harder	checked	checked
Cross-modal	harder	checked	checked
Error handling	checked	checked	checked

Pass criteria

1. Correct handoff sequence
2. Successful completion
3. Output artifact / answer
4. Graceful errors for invalid requests

Automatic test evaluation model completely redesigned and updated

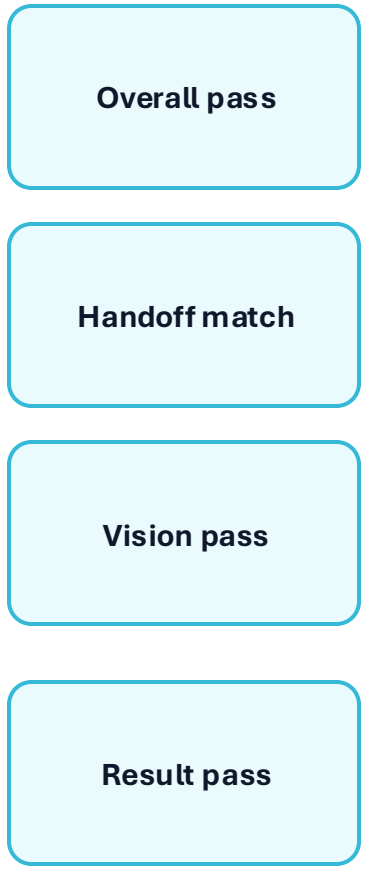
Evaluation metrices

- Overall pass
- Handoff match
- Vision pass
- Result pass
- Tool-call efficiency
- Robustness by prompt level
- Throughput tokens/sec
- Tokens/task
- Energy/run
- Avg latency
- TTFT
- Passes per min accuracy ÷ time
- Passes per kWh accuracy ÷ energy
- Error Handling

METRICS

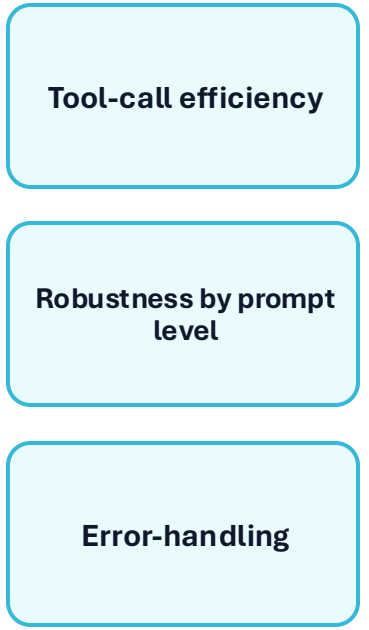
Correctness — "did it get the task right?"

- **Overall pass**: End-to-end success across all 66 runs.
- **Handoff match**: Correct agent routing across all 66 runs.
- **Vision pass**: Vision/VQA answer quality, measured only on scenarios where vision_pass is defined.
- **Result pass**: Ground-truth answer correctness, measured only on scenarios where `result_pass` is defined.



Behavioral quality — "how did it work?"

- **Tool-call efficiency** — Redundant agent calls beyond the minimal correct route.
- **Error-handling** — Correct rejection/clarification on invalid or impossible tasks.
- **Prompt-level robustness** — Pass rate split by vague, clear, and algorithmic prompts.



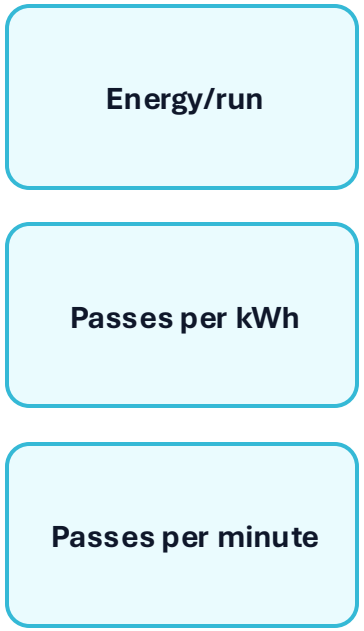
Speed and Cost

- **Avg latency** — End-to-end user wait time per task.
- **TTFT** — Time until the model starts streaming tokens.
- **Throughput (tok/s)** — Token generation speed after streaming begins.
- **Tokens/task** — Average total LLM tokens used per scenario.



Energy & Efficiency

- **Energy/run** — Estimated electricity used by the full 66-run suite.
- **Passes per kWh** — Correct answers per unit of energy.
- **Passes per minute** — Correct answers per minute of wall-clock time.

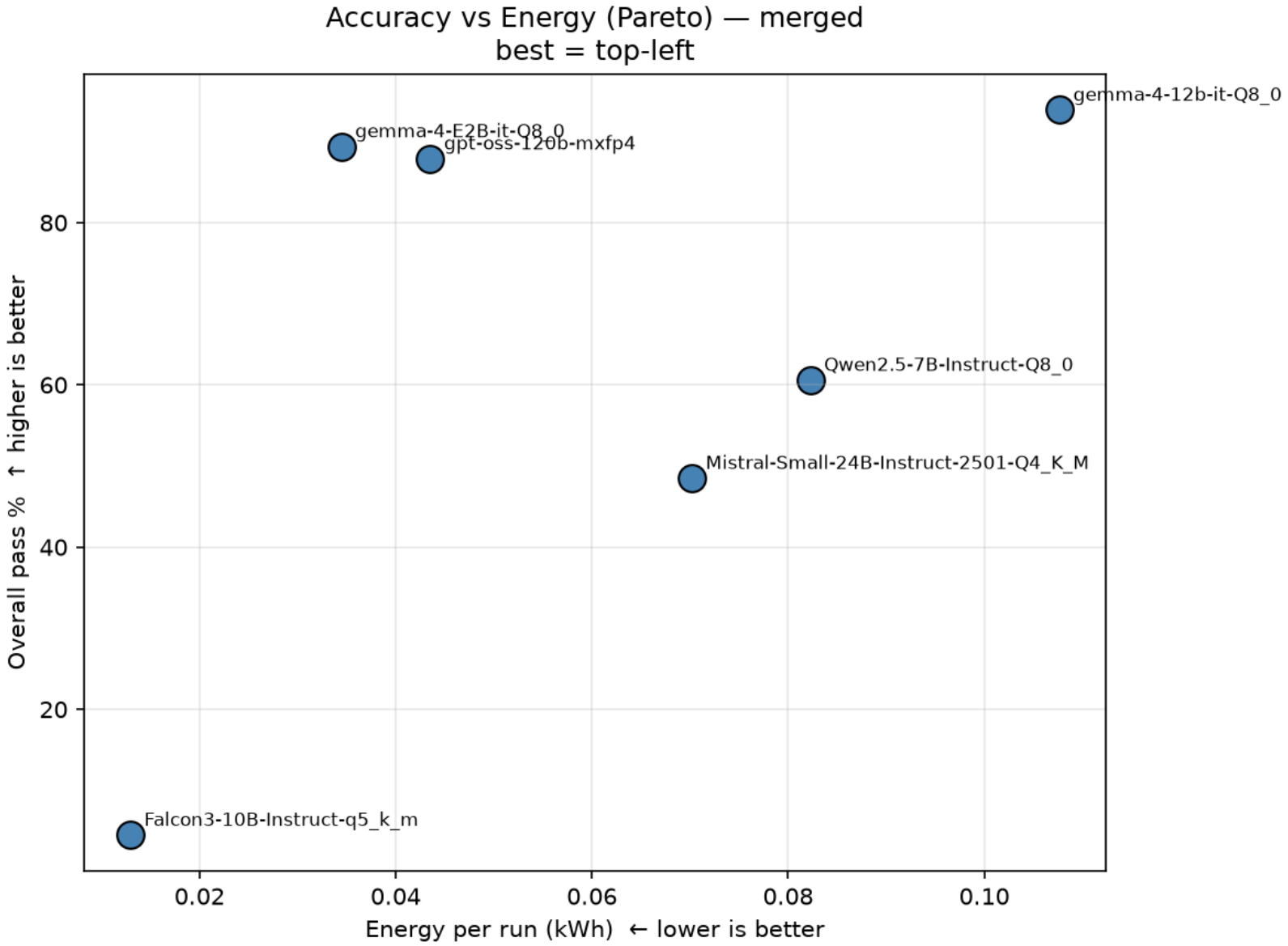


METRICS

Results

Metric	gemma-12b	gemma-E2B	gpt-oss-120b	qwen2.5-7B	mistral-24B	falcon3-10B
Country	USA	USA	USA	China	France	UAE
Size · Arch	12 d	5 d	120 MoE	7 d	24 d	10 d
1 · Overall %	**93.9**	89.4	87.9	60.6	48.5	4.5
2 · Handoff %	95.5	92.4	87.9	72.7	56.1	15.2
3 · Vision %	100.0	77.8	88.9	100.0	66.7	66.7
4 · Result %	92.6	92.6	96.3	70.4	53.8	22.2
5 · Tool-call eff. (extra/task)	0.10	0.20	0.07	1.17	0.24	0.20
6 · Error-handling %	100.0	100.0	92.9	64.3	57.1	14.3
7a · Robust vague %	90.9	77.3	72.7	50.0	45.5	0.0
7b · Robust clear %	90.9	95.5	95.5	68.2	54.5	4.5
7c · Robust algo %	100.0	95.5	95.5	63.6	45.5	9.1
8 · Avg latency (s)	106.50	36.70	41.30	90.70	67.20	16.20
9 · TTFT (ms)	397.90	91.80	306.00	82.40	574.50	93.80
10 · Throughput (tok/s)	16.2	78.4	52.7	28.2	14.1	26.7
11 · Tokens/task	14,945	15,248	12,252	20,632	16,950	12,830
12 · Energy/run (kWh)	0.108	0.034	0.044	0.082	0.070	0.013
13 · Passes/kWh	576.00	**1712.00**	1333.00	486.00	456.00	232.00
14 · Passes/min	0.53	1.46	1.28	0.4	0.43	0.17

Results


Accuracy vs Energy (Pareto) — merged
best = top-left
gemma-4-12b-it-Q8_0
gemma-4-E2B-it-Q8_0
gpt-oss-120b-mxfp4
Qwen2.5-7B-Instruct-Q8_0
Mistral-Small-24B-Instruct-2501-Q4_K_M
Falcon3-10B-Instruct-q5_k_m
Overall pass % ↑ higher is better
80
60
40
20
0.02
0.04
0.06
0.08
0.10
Energy per run (kWh) ← lower is better

WRAP-UP

Conclusion and next steps

Vision-aware VideoSearcher

The system now reasons over what appears in video frames, not only over transcripts.

Cross-modal reasoning

Spoken phrases can become timestamps, frames and visual answers.

Modular and testable

The extension preserves MCP modularity and adds a repeatable evaluation path.

Next steps

UI/chat interface · segmentation benchmarks · RAG over transcripts · LLM as a judge · Memory